Noémi Kreif

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BIOSKETCH

I am an **Assistant Professor of Health Economics** at the CHOICE Institute in the Department of Pharmacy at the University of Washington School of Pharmacy, where I joined the faculty in March 2024. My scholarship centers on the development and application of rigorous **causal inference methods** for evaluating health policies and interventions, **integrating modern machine learning approaches** with econometric and epidemiologic methods to estimate heterogeneous treatment and policy effects to better inform targeting, personalization, and policy design. I have applied these methods across a wide range of **substantive areas**, including **national health insurance reforms**, **provider payment policies**, digital health interventions, **mental health** and economic outcomes, and the health impacts of conflict and violence in both **high-income** and **low- and middle-income** country **settings**. I received my PhD in Health Economics from the London School of Hygiene and Tropical Medicine (University of London) in 2013 and previously held postdoctoral and faculty positions at LSHTM and the University of York, where I was promoted to Senior Research Fellow (equivalent to Associate Professor). I have authored over **50 peer-reviewed journal articles** and invited chapters (h-index 27, Google Scholar) across health economics, health policy, epidemiology, and statistics. I served as principal or co-investigator on grants funded by major agencies including the UK Medical Research Council, NIHR, the Wellcome Trust, and the European Commission.

CURRENT POSITION

2024-	**Assistant Professor,** The Comparative Health Outcomes, Policy, and Economics (CHOICE) Institute, School of Pharmacy, University of Washington

QUALIFICATIONS

2013	**PhD in Health Economics,** London School of Hygiene and Tropical Medicine (LSHTM), University of London (UK). Thesis: "Statistical methods to address selection bias in economic evaluation that uses observational data"
2008	**MA in Economics,** Central European University, Hungary
2006	**MSc in Economics**, Corvinus University of Budapest, Hungary; Rajk Laszlo College for Advanced Studies, Hungary

PAST POSITIONS

2021 - 2024	**Senior Research Fellow (Associate Professor) in Health Economics** University of York Centre for Health Economics **(CHE)**
2016 - 2021	**Research Fellow (Assistant Professor)** in Health Economics University of York, CHE
2015 - 2016	**Lecturer (Assistant Professor)** in Health Economics, LSHTM, Medical Research Council Early Career Fellow in the Economics of Health (2014-2017)
2013 - 2015	**Postdoctoral Research Fellow** in Health Economics, LSHTM
2006 - 2012	**Research Associate** in Health Economics at United Biosource Corporation

VISITING POSITIONS

2022	**Visiting Researcher,** Center for Targeted Machine Learning and Causal Inference, UC Berkeley
2019-	**Honorary Research Fellow** at the Health Economics and Health Technology Assessment (HEHTA) at the **University of Glasgow**
2016	**Visiting Researcher,** Kaiser Permanente Division of Research
2015	**Visiting Researcher** at the Division of Biostatistics, UC Berkeley
2010	**Visiting Graduate student** at the Center for Causal Inference and Program Evaluation, UC Berkeley

RESEARCH GRANT INCOME (AS PRINCIPAL OR CO-INVESTIGATOR)

1. **Kreif N** (PI): MQ Foundation Fellowship in Economics and Mental Health Research ($100,000, Kennedy Forum. 9/1/2026-8/31/2027
2. Suhrcke M (PI), Moreno Serra R,…, **Kreif N,** Muzumdar S, Walker S,**…:** NIHR Global Health Research Unit on "Health financing on UHC in Challenging times: leaving no-one behind). £6,987,000, 01/02/2022 → 31/01/2027 (**participation until 2024 March**)
3. Tiffin P (PI), **Kreif N**, Paton L, Aylott L: Developing a Rapid AI-based Policy Probing and Observational Research Tool. Wellcome Data Prize (Round 1) £40,000, 01/09/2022 → 01/03/2023
4. Moreno Serra R (PI), **Kreif, N**. & Lordemus, S.UKRI GCRF/Newton Fund - COVID-19, social distancing and violence against women in Brazil (BRAVE). MEDICAL RESEARCH COUNCIL: £107,406.4021/08/20 → 20/02/22
5. **Kreif, N (PI)**, James Cussens: Tailoring health policies to improve outcomes using machine learning, causal inference and operations research methods (MEDICAL RESEARCH COUNCIL NIRG: £407,366. 01/09/20 → 01/09/23)
6. Jacobs, R (PI), **Kreif, N**, Mirelman, A, Suhrcke, M, Villasenor-Lopez, A: The causal impact of mental health on economic outcomes: evidence from a Mental Health Intervention for Chronic Disease Patients (MIND_ECON) (MEDICAL RESEARCH COUNCIL: £137,965.47, 1/01/19 → 31/12/21)
7. Manca, A (PI), Iglesias Urrutia, C. P., **Kreif, N**., Rothery, C., Smith, A. G. & Yu, G. HTx: Next Generation HTA (EUROPEAN COMMISSION: £778,587.75, 1/01/19 → 31/12/23)
8. Moreno Serra, R. (PI), Caspersen, N., **Kreif, N**., Mirelman, A. & Suhrcke: War and Peace: The Health Consequences of Conflict in Colombia (MEDICAL RESEARCH COUNCIL: £379,068.00, 1/04/18 → 30/09/20)
9. Suhrcke, M. (PI), Chalkley, M. J., Cookson, R. A., **Kreif, N**., Mirelman, A., Moreno Serra, R., Palmer, S. J., Revill, P., Rice, N. & Sculpher, M.: GHE2 - The Economics and Econometrics of Global Health Systems, Programmes and Policies (NIHR GHRF, £1,111,624.00, 1/08/17 → 31/07/20)
10. **Kreif, N**. (PI) Improving statistical methods to address confounding in the economic evaluation of health interventions (MEDICAL RESEARCH COUNCIL: £251,171, 9/05/14 → 9/05/17)
11. PhD studentship from the Economic and Social Research Council (UK), inclusive of the Advanced Quantitative Methods Enhanced Stipend, 2009-2012

PUBLICATIONS (*: FIRST/JOINT FIRST/SENIOR AUTHOR)

Citations: 3056, h-index: 28 (Google Scholar)

INVITED EDITORIALS, NOTES, PEER REVIEWED BOOK CHAPTERS

1. *Hatamyar J, **Kreif, N** (forthcoming). Machine Learning for Causal Inference: Opportunities and Pitfalls."In the Oxford Research Encyclopedia of Economics and Finance. Oxford University Press. doi:[#]
2. Powell-Jackson T, O'Neill S, **Kreif N.** Difference-in-differences. In: *Evaluating Public Health Interventions (textbook)*, University of College Press, 2026 (in press)
3. ***Kreif N**. Learning from an association analysis using propensity scores. *Pediatric Critical Care Medicine*. 2021 Dec 1;22(12):1088-92.
4. Shah V, **Kreif N** and Jones AM. Machine Learning for Causal Inference: Estimating Heterogeneous Treatment Effects. *In: Edward Elgar Handbook in Empirical Microeconomics* 2021.
5. ***Kreif N**, DiazOrdaz K. Machine learning in policy evaluation: new tools for causal inference, In: *Oxford Research Encyclopedia of Economics and Finance*. 2019

PEER REVIEWED JOURNAL ARTICLES

1. Ochoa-Moreno I, **Kreif N**, Hidayat T, Kurnia S, Hidayat B, Griffin S. Does increasing overall health expenditure reduce inequality in under 5 mortality rates between provinces in Indonesia? *Health Policy and Planning*. 2026 Jun 18:czag081.
2. *Kneitel R, Lee K, **Kreif N.** Health care resource utilization and costs in patients with acute myeloid leukemia treated with posttransplant maintenance therapy. *Journal of Managed Care & Specialty Pharmacy*. 2026 Aug;32(8):932-44.
3. Shah V, Hatamyar J, Hidayat T, **Kreif N**. Exploring the heterogeneous impacts of Indonesia's conditional cash transfer scheme (PKH) on maternal health care utilisation using instrumental causal forests. *Empirical Economics*. 2026 Feb;70(2):39.
4. Rodrigues D, **Kreif N**, Darzi A, Barahona M, Mayer E. Key factors behind the adoption of a patient-facing technology in primary care: a structured approach to variable importance. *Health Services and Outcomes Research Methodology*. 2025 Oct 25:1-9.
5. Kalo Z, Petrova G, Németh B, Došenović Bonča P, Delnoij D, Bennett A, Tachkov K, Zhang Y, **Kreif N**, Knies S, Lipska I. Transferability of Real World Evidence to Support HTA Recommendations in Lower Income European Countries. *Health Science Reports*. 2025 Oct 13.
6. *Zhang Y, Bennett A, Manca A, Mittelman M, Hoeks M, Smith A, Taylor A, Stauder R, de Witte T, Malcovati L, van Marrewijk C, **Kreif N**. Estimating the Causal Effect of Realistic Treatment Strategies Using Longitudinal Observational Data. *Medical Decision Making*. 2025 Aug 2:0272989X251379819.
7. *Shah V, Jones AM, Malenica I, Hidayat T, **Kreif N**. Using policy learning to inform health insurance targeting: a case study of Indonesia. *Health Economics*. 2025 Sep 8. https://doi.org/10.1002/hec.70031
8. *Shah V, **Kreif N.** Estimating Heterogeneous Impacts Of Subsidised Health Insurance: A Causal Machine Learning Approach. *Plos One* https://doi.org/10.1371/journal.pone.0315057
9. Rodrigues D, **Kreif N** Barahona, M, Mayer, Erik.: Digitalisation of access to primary care: is there an equity-efficiency trade-off? *Health Economics* https://doi.org/10.1002/hec.70014
10. Paton LW, **Kreif N**, Aylott L, Kerrigan P, Nader C, Gega L, Tiffin PA: The impact of physical activity on psychological distress in young people: an analysis of longitudinal data from the Millenium Cohort Study using causal machine learning. *JCPP Advances* https://doi.org/10.1002/jcv2.70035
11. Diaz M, Pereda P, Rocha F, Oliveira P, Arabe I, **Kreif N**, Lordemus S, Moreno-Serra R: Public Policies and Femicides during the COVID-19 Pandemic. *Economics and Human Biology* 2025. Sep:58:101508.

12. *McGuire F, Santos R, Smith PC, Stacey N, Edoka I, **Kreif N**. Health facility quality peer effects: Are financial incentives necessary?. *Regional Science and Urban Economics*. 2025 May 1;112:104091.
13. Counts NZ, **Kreif N**, Creedon TB, Bloom DE, Psychological Distress in Adolescence and Later Economic and Health Outcomes in the US Population. *PLoS Medicine.* 2025. 22 (1), e1004506
14. * McGuire F, Smith PC, Stacey N, Edoka I**, Kreif N**. Do health care quality improvement policies work for all? Distributional effects by baseline quality in South Africa. Health Economics. 2025 Jan;34(1):175-99.
15. Zhang Y, **Kreif N**, Gc VS, Manca A. Machine Learning Methods to Estimate Individualized Treatment Effects for Use in Health Technology Assessment. *Medical Decision Making*. 2024 Oct;44(7):756-69.
16. *Glynn, D., Giardina J, Hatamyar, J**,** Pandya A, **Kreif N.** Integrating decision modeling and machine learning to inform treatment stratification. *Health Economics* DOI: 10.1002/hec.4834
17. Lordemus S, **Kreif N**, Moreno-Serra R. Public healthcare financing during counterinsurgency efforts: Evidence from Colombia. *Oxford Bulletin of Economics and Statistics*. 86 (5), 1230-1259
18. Salas-Ortiz A, Moreno-Serra R, **Kreif N**, Suhrcke M, Casas G. The effect of conflict-related violence intensity and alcohol use on mental health: The case of Colombia. *Social Science in Medicine-Population Health*. 2024 Feb 8:101626.
19. Rocha F, Diaz MD, Pereda PC, Árabe IB, Cavalcanti F, Lordemus S, **Kreif N**, Moreno-Serra R. COVID-19 and violence against women: Current knowledge, gaps, and implications for public policy. *World Development*. 2024 Feb 1;174:106461.
20. *S Moler-Zapata, **N Kreif**, J Ochalek, AJ Mirelman, M Nadjib, M Suhrcke: Estimating the Health Effects of Expansions in Health Expenditure in Indonesia: A Dynamic Panel Data Approach, *Applied Health Economics and Health Policy*, 1-11, 2022
21. D Rodrigues, **N Kreif**, A Lawrence-Jones, M Barahona, E Mayer: Reflection on modern methods: constructing directed acyclic graphs (DAGs) with domain experts for health services research. *International Journal of Epidemiology* 51 (4), 1339-1348, 2022
22. WV Padula, **N Kreif**, DJ Vanness, B Adamson, JD Rueda, F Felizzi, ...Machine Learning Methods in Health Economics and Outcomes Research—The PALISADE Checklist: A Good Practices Report of an ISPOR Task Force, *Value in Health* 25 (7), 1063-1080, 2022
23. MJ Aragón, M Chalkley, **N Kreif**: The long-run effects of diagnosis related group payment on hospital lengths of stay in a publicly funded health care system: Evidence from 15 years of micro data, *Health Economics* 31 (6), 956-972, 2022
24. D Rodrigues, **N Kreif**, K Saravanakumar, B Delaney, M Barahona, ...Formalising triage in general practice towards a more equitable, safe, and efficient allocation of resources. *BMJ* 377, 2022
25. G Ciminata, C Geue, O Wu, M Deidda, **N Kreif,** P Langhorne. Propensity score methods for comparative-effectiveness analysis: A case study of direct oral anticoagulants in the atrial fibrillation population. *PloS One* 17 (1), e0262293, 2022
26. ***Kreif N**, DiazOrdaz K, Moreno-Serra R, Mirelman A, Hidayat T, Suhrcke M. Estimating heterogeneous policy impacts using causal machine learning: a case study of health insurance reform in Indonesia. *Health Services and Outcomes Research Methodology*. 2021 Nov 9:1-36.
27. ***Kreif N**, Mirelman A, Suhrcke M, Buitrago G, Serra RM. The impact of civil conflict on child health: evidence from Colombia. *Economics & Human Biology*. 2021 Oct 28:101074.
28. McGuire F, **Kreif N**, Smith PC. The effect of distance on maternal institutional delivery choice: Evidence from Malawi. *Health Economics*. 2021 Jun 6.
29. León-Giraldo S, Casas G, Cuervo-Sánchez JS, González-Uribe C, Olmos A, **Kreif N**, Suhrcke M, Bernal O, Moreno-Serra R. A light of hope? Inequalities in mental health before and after the peace agreement in Colombia: a decomposition analysis. *International Journal for Equity in Health*. 2021, 20: (39)

30. Leon-Giraldo S, Cuervo-Sanchez JS, Casas G, González-Uribe C, **Kreif N**, Bernal O, Moreno-Serra R. Inequalities in Catastrophic Health Expenditures in Conflict-Affected Areas and the Colombian Peace Agreement: An Oaxaca Blinder Change Decomposition Analysis. *International Journal for Equity in Health*, 29 Sep 2021, 20(1):217
31. Stacey N, Edoka I, Suhrcke M, **Kreif N**, Mirelman A, Hoffman, K: Facility standards and the quality of public sector primary care: Evidence from South Africa's "Ideal Clinics" *Health Economics*. First published: 17 March 2021 https://doi.org/10.1002/hec.4228
32. ***Kreif N**, Mirelman A, Love-Koh J, Kim S, Moreno Serra R, Revill P, Sculpher M, Suhrcke M From impact evaluation to decision-analysis: assessing the extent and quality of evidence on 'value for money' in health impact evaluations in low- and middle-income countries. *Gates Open Research* ([version 1; peer review: 1 approved], 2021
33. Chi, YL Blecher M, Chalkidou K Culyer A, Klaxton K, Edoka I, Glassman A, **Kreif N** et al. What next after GDP-based cost-effectiveness thresholds? *Gates Open Research* ([version 1; peer review: 2 approved], 2021
34. ***Kreif N**, Sofrygin O, Schmittdiel J, Adams A, Grant R, Zhu Z, van der Laan M, Neugebauer R. Exploiting non-systematic covariate monitoring to broaden the scope of evidence about the causal effects of adaptive treatment strategies. *Biometrics*, 2020 First published: 15 April 2020 https://doi.org/10.1111/biom.13271
35. O'Neill S, **Kreif N,** Sutton M, Grieve R. A comparison of methods for health policy evaluation with controlled pre-post designs. *Health Services Research*. 2020 Apr;55(2):328-38.
36. Gaughan J, Gutacker N, Grašič K, **Kreif N,** Siciliani L, Street A. Paying for efficiency: incentivising same-day discharges in the English NHS. *Journal of Health Economics*. 2019 Dec 1;68:102226.
37. Deidda M, Geue C, **Kreif N**, Dundas R, McIntosh E. A framework for conducting economic evaluations alongside natural experiments. *Social Science & Medicine*, 2019, 220(353-361)
38. DiazOrdaz K, Daniel R, **Kreif N**. Data-adaptive doubly robust instrumental variable methods for treatment effect heterogeneity. *The Journal de la Société Française de Statistique*, special issue on Causality, 135-163. ISSN 2102-6238
39. ***Kreif N**, Tran L, Grieve R, Tasker R, Petersen M Comparing longitudinal treatment regimes in critical care: an illustration of semi-parametric methods, *American Journal of Epidemiology.* 2017, 12(15),1370-1379
40. O'Neill S, **Kreif N**, Grieve R, Sutton, M Sekhon JS, Estimating causal effects: considering three alternatives to difference-in-differences estimation. Health Services and Outcomes Research Methodology. 2016, 1, 1–21
41. ***Kreif N,** Grieve R, Hangartner D, Turner AJ, Nikolova S, Sutton M. Examination of the Synthetic Control Method for Evaluating Health Policies with Multiple Treated Units. *Health Economics*, 2016, 25, 1514–1528
42. ***Kreif N,** Grieve R, Díaz I, Harrison D. Evaluation of the Effect of a Continuous Treatment: A Machine Learning Approach with an Application to Treatment for Traumatic Brain Injury. *Health Economics*, 2015, 24(9), 1213-1228.
43. Miners A, Phillips A**, Kreif N**, Rodger A, Speakman A, Fisher M. et al. Health-related quality-of-life of people with HIV in the era of combination antiretroviral treatment: a cross-sectional comparison with the general population. *The Lancet HIV*, 2014, 1(1), e32-e40.
44. ***Kreif N**, Grieve, R., Radice R ., Gruber S, Sekhon JS. Evaluating treatment effectiveness under model misspecification: A comparison of targeted maximum likelihood estimation with bias-corrected matching. *Statistical Methods in Medical Research***,** 2016; 25(5): 2315–2336.

45. ***Kreif N,** Grieve R, Radice R, Sekhon JS. Regression-adjusted matching and double-robust methods for estimating average treatment effects in cost-effectiveness analyses. *Health Services and Outcomes Research Methodology*, 2013, 13(2-4), 174-202.
46. Ishak, KJ., **Kreif N**., Benedict A, Muszbek N. Overview of Parametric Survival Analysis for Health-Economic Applications. *PharmacoEconomics*, 2013, 31(8):663-75
47. ***Kreif N,** Grieve R, Sadique ZM. Statistical methods for cost-effectiveness analyses that use observational data: a critical appraisal tool and review of current practice. *Health Economics*, 2013, 22(4):486-500.
48. ***Kreif N,** Grieve R, Sadique ZM, Radice R, Ramsahai R, Sekhon JS. Methods for estimating subgroup effects in cost-effectiveness analyses that use observational data. *Medical Decision Making*, 2012, 32(6):750-63.
49. Muszbek N, **Kreif N**, Valderrama A, Benedict A, Ishak J, Ross P. Modeling Survival in Hepatocellular Carcinoma. *Current Medical Research and Opinion*, 2012, 28(7), p 1141-53.
50. Radice R**,** Ramsahai R, Grieve R, **Kreif N**, Sadique ZM, Sekhon JS: Evaluating treatment effectiveness in patient subgroups: a comparison of propensity score methods with an automated matching approach. *International Journal of Biostatistics,* 2012, 7;8(1):25.
51. Calvo Aller E, Rey PM **Kreif N**, et al.: Cost-effectiveness evaluation of sunitinib as first-line targeted therapy for metastatic renal cell carcinoma in Spain. *Clin and Transl Onc*, 2011, 13(12), p 869-877.
52. Lux MP, Wöckel A, Benedict A, Buchholz S, **Kreif N**, et al.: Cost-effectiveness analysis of anastrozole versus tamoxifen in adjuvant therapy for early-stage breast cancer - a health-economic analysis based on the 100-month analysis of the ATAC trial and the German health system. *Onkologie*, 2010, 33(4),p 155-66.
53. Järbrink K, **Kreif N**, Benedict A, Locklear J: Quality of life and drug costs associated with switching antipsychotic medication to once-daily extended release quetiapine fumarate in patients with schizophrenia. *Current Medical Research and Opinion,* 2009, 25(3),p 709-716(8).

WORKING PAPERS IN THE PUBLIC DOMAIN

1. Cussens J, Hatamyar J, Shah V**, Kreif N**. Fast Learning of Optimal Policy Trees. arXiv preprint arXiv:2506.15435. 2025 Jun 18. (paper submitted to Journal of Machine Learning Software)
2. Hatamyar J, **Kreif N**, Rocha R, Huber M. Machine Learning for Staggered Difference-in-Differences and Dynamic Treatment Effect Heterogeneity. arXiv preprint arXiv:2310.11962. 2023 Oct 18.
3. *Hatamyar J, **Kreif N**. Policy Learning with Rare Outcomes. arXiv preprint arXiv:2302.05260. 2023 Feb 10.
4. *Shah V, Hatamyar J, Hidayat T, **Kreif N**. Exploring the heterogeneous impacts of Indonesia's conditional cash transfer scheme (PKH) on maternal health care utilisation using instrumental causal forests. arXiv preprint arXiv:2501.12803. 2025 Jan 22.
5. ***Kreif N**, Mirelman A, Moreno Serra R, Hidayat T, DiazOrdaz K, Suhrcke M. Who benefits from health insurance?: Uncovering heterogeneous policy impacts using causal machine learning. CHE Research Paper 17, University of York. 2020 (Published in *Health Services and Outcomes Research Methodology*)
6. ***Kreif N**, Sofrygin O, Schmittdiel J, Adams A, Grant R, Zhu Z, van der Laan M, Neugebauer R. Evaluation of adaptive treatment strategies in an observational study where time-varying covariates are not monitored systematically. arXiv preprint arXiv:1806.11153. 2018 Jun 28. (Published in *Biometrics*)

DRAFT MANUSCRIPTS PRE- SUBMISSION OR UNDER REVISION

Causal machine learning for health policy evaluation:

1. Policy Learning in Practice: Simulation Evidence and a Case Study of Optimal Allocation of Subsidised Health Insurance (Hatamyar J, **Kreif N**) (Under Revision with *Statistics in Medicine*)
2. "Comparative Evaluation of Machine Learning Methods for Personalized Treatment Allocation based on Randomized Controlled Trial Data" (**Kreif N**, Basu A)
3. "Machine learning in differences in differences with staggered treatment timing: a case study of a primary health care reform in Brazil (working title) (Hatamyar J, **Kreif N**, Huber M, Rocha R)

Causal inference methods for decision science:

4. "Target Trial Emulation (TTE) for real world data analyses to support HTA decisions" (Bennett A, **Kreif N**, Manca A)

Global health policy:

5. "A Solution Looking for a Problem: The Effect of Maternity Waiting Homes on Health Care Utilisation and Child Health Outcomes" (McGuire F, **Kreif N**, Chirwa G, Manthalu G, Smith P)
6. "The Impact of Mental Health Status on Employment, Income, Out Of Pocket Payments for Healthcare and Household Spending Patterns" (Jacob N, **Kreif N**, Susan Lordemus S, Cleary S, Jacobs R) (Submitted to *The Journal of Population Economics*)
7. "Does increasing overall health expenditure reduce inequality in under 5 mortality rates between provinces in Indonesia?" (Ochoa I, Gruber S, **Kreif N**) (Under Revision with Health Policy and Planning)

Applications in health economics and outcomes research

8. "The promise and challenges for machine learning in analysing real world data for health technology assessment" (**Kreif N**, Manca A, Salanti G, Zhang Y, Bennett A, Sirthola P et al.)
9. "Healthcare Resource Utilization and Costs Associated with Depression Among People with Alzheimer's Disease - A Retrospective Cohort Study" (O Yip, C. English, **Kreif N**, K. Lee)
10. **"**Real-World Comparative Effectiveness of Abiraterone, Apalutamide, Enzalutamide, and Docetaxel Doublets for Metastatic Hormone-Sensitive Prostate Cancer" (Nwogu IB , **Kreif N** Khan H, Etzioni R, Carlson J)

PUBLISHED CONFERENCE ABSTRACTS

1. Nwogu IB, **Kreif N**, Khan HM, Etzioni RD, Carlson JJ. Real-world comparative effectiveness of novel doublet therapies for metastatic hormone-sensitive prostate cancer. ASCO 2026
2. Kneitel R, **Kreif N**. EE84 Cost-Effectiveness Analysis of Xanomeline and Trospium Chloride for Schizophrenia in the United States. Value in Health. 2025 Jul 1;28(6):S77.
3. Gallinaro J, Shah V, Lee H, **Kreif N**, Guerra I. P23 Integrating Machine Learning and Propensity Score-Based Methods for Making Causal Inferences From Real-World Data. Value Health. 2024;27(12):S6.
4. Glynn D, Hatamyar J, Giardina J, Pandya A, **Kreif N**. Identifying equity constrained subgroups using machine learning, decision modelling and optimal policy algorithms. Med Decis Making. 2024;44(2):NP193-NP196.
5. Glynn D, **Kreif N**, Hatamyar JF, Pandya A, Giardina J. Integrating machine learning estimates of heterogeneous treatment effects and decision modelling. In: Society for Medical Decision Making (SMDM), Seattle 2022.
6. Bennett A, **Kreif N**, Manca A. MSR52 Target Trial Emulation (TTE) for Real World Data Analyses to Support HTA Decisions. Value Health. 2022;25(12):S359-S360.

7. Zhang Y, **Kreif N**, Gc V, Manca A. MSR53 A Review of Methods for Estimating Individual Treatment Effect From Real World Data for Use in Health Technology Assessment: Separating Hype From Reality. Value Health. 2022;25(12):S360.
8. Zhang Y, **Kreif N**, Gc V, Bennett A, Manca A. EE147 Estimating the Causal Effect of Early Use of Erythropoietic Stimulating Agents in Intermediate-1 to Low-Risk MDS Patients: An Application of the Longitudinal Targeted Maximum Likelihood Estimation. Value Health. 2022;25(12):S81.
9. Jacob N, **Kreif N**, Lordemus S, Suhrcke M, Jacobs R, Myers B, Mafunda V, Cleary S. The causal impact of mental health on economic outcomes in South Africa: employment, income, out of pocket payments for healthcare, and household expenditure. J Ment Health Policy Econ. 2022 Mar 1;25:S17. INT CTR MENTAL HEALTH POLICY & ECONOMICS-ICMPE.
10. Diaz MD, Rocha F, Pereda P, **Kreif N**, Lordemus S, Moreno-Serra R. Evidence from Developing Countries on COVID-19 Lockdowns and Violence Against Women: A Literature Review. World Congress on Health Economics 2021 Jul 12. IHEA.
11. Pereda P, Diaz MD, Rocha F, **Kreif N**, Lordemus S, Moreno-Serra R, Arabe IB, Oliveira PA, Cavalcanti F. COVID-19 Lockdowns and Violence Against Women in Brazil. World Congress on Health Economics 2021 Jul 12. IHEA.
12. Ciminata G, Geue C, Wu O, Deidda M, **Kreif N**, Langhorne P. PND96 COMPARISON OF PROPENSITY SCORE METHODS A CASE STUDY OF DIRECT ORAL ANTICOAGULANTS. Value Health. 2019;22:S755.
13. Lampe F, Miners A, **Kreif N**, Miltz A, Phillips A, Sewell J. Comparison of depression and anxiety between HIV-positive and HIV-negative people. 22nd International AIDS Conference; 2018; Amsterdam.
14. Ciminata G, Geue C, Wu O, Deidda M, **Kreif N**, Langhorne P. Comparative Effectiveness of Non-Vitamin K Antagonist Oral Anticoagulants (NOACS) And Warfarin In The Scottish Atrial Fibrillation Population: The Value of Real World Evidence. Value Health. 2017;20(9):A603..
15. Stull D, Wasiak R, **Kreif N**, Colligs A, Seitz C, Gerlinger C. PIH28 VALIDATION OF THE SF-36 IN PATIENTS WITH ENDOMETRIOSIS. Value Health. 2009;12(7):A295.
16. Wasiak R, Filonenko A, Stull DE, **Kreif N**, Raluy M, Ryan J, Jeddi M, et al. PIH40 Methodological considerations when assessing work productivity (WP) and activities of daily living (ADL) outcomes in multinational clinical trials in women with heavy and … Value Health. 2010;13(7):A382.
17. Wasiak R, **Kreif N**, Stull D, Tyas DA. PCV79 Burden of illness study in patients with resistant hypertension in UK. Value Health. 2009;12(7):A327.
18. **Kreif N**, Benedict Á, Lux MP, Wöckel A, Klevesath MB. PCN119 Cost effectiveness analysis of anastrozole versus tamoxifen in adjuvant therapy for early stage breast cancer based on the 100-month analysis of the ATAC trial. Value Health. 2009;12(7):A279-A280.
19. Stull D, Wasiak R, **Kreif N**, Colligs A, Seitz C, Gerlinger C. Validation of the SF-36 in patients with endometriosis. Value Health. 2009 Oct;12(7):A295.
20. Benedict A, Figlin RA, Charbonneau C, **Kreif N**, Hariharan S, Négrier S. Economic evaluation of sunitinib versus other new targeted therapies as first-line treatment of metastatic renal cell carcinoma (mRCC) in the United States. Journal of Clinical Oncology. 2009 May 20;27(15_suppl):e17556-.

INVITED SPEAKER AT SEMINARS AND CONFERENCES (TITLES FROM BEFORE 2016 OMITTED)

1. **Stanford Department of Health Policy Seminar series (**Stanford, California, November 2026)
    a. Title TBA

2. **PriMHE - Programme in the Methods of Health Economics Lecture series, University of Vienna, , September 2026** (invited online lecture)
    a. "(Causal) Machine Learning for Decision modelling"
3. **Frontiers in Causal Modeling: SMDM – ISPOR Panel (invited panelist), SMDM 48th Annual Meeting** (Oslo, Norway)
    - "(Causal) Machine Learning for Decision modelling"
4. **Rajk College Neumann Jubilee Conference (invited keynote speaker)**, Corvinus University (Budapest, Hungary, 2026)
    • "How does the work of Neumann Leuarates Inform Data Driven Health Policy Evaluation?"
5. **Tufts Center for the Evaluation of Value and Risk in Health (invited seminar speaker, virtual, May 2025)**
    • "Informing treatment stratification using causal machine learning and decision modelling"
6. **Skaggs School of Pharmacy and Pharmaceutical Sciences (invited seminar speaker) (Colorado, March 2025)**
    • "Informing treatment stratification using causal machine learning and decision modelling"
7. **Spanish Health Economics Association Conference (Tenerife, Spain, 2024) (speaker in invited session)**
    • "Policy learning with rare outcomes"
8. **German Health Economic Association (invited seminar presentation)**
    • "Machine learning in health economics and outcomes research: opportunities and some key challenges"
9. **Horizontal Initiative on Causal Inference of the German Society for Epidemiology (Mainz, Germany, November 2023) (invited keynote speaker)**
10. **CHOICE (University of Washington) annual symposium: Artificial Intelligence, Real Consequences: The side effects of AI in healthcare (Seattle, USA, 2023 October) (panelist)**
    • "Machine learning to answer causal questions: how to avoid misleading interpretation?"
11. **Hungarian Economic Association Conference keynote lecture (Hungary, December 2022)**
    • "Data driven health policy making"
12. **Harvard Centre for Health Decisions Science Seminar (Boston, USA, October 2022)**
    • "Learning Optimal Treatment Rules Using Machine Learning and Decision Modelling"
13. **University of East Anglia Economics Department Seminar (UK, 2022)**
    • "Policy Learning with rare outcomes"
14. **Stanford AI + Health Conference (2021) (panelist)**
    • "Machine learning for causal inference (in health economics and beyond)"
15. **Virtual ISPOR Europe Conference (2021) (panelist)**
    • "Machine learning for causal inference"
16. **Rajk Laszlo College Alumni Conference, Budapest (Hungary 2021)**
    • "Using machine learning in the impact evaluation of health policies"
17. **Nurenberg Economics Seminar (Germany 2020)**
    • "Who benefits from health insurance? Uncovering heterogeneous policy impacts using causal machine learning"
18. **Essen Virtual Health Economics Seminar (Germany 2020)**
    • "Who benefits from health insurance? Uncovering heterogeneous policy impacts using causal machine learning"
19. **Royal Statistical Society Interpretable Machine Learning & Causal Inference workshop (December 2020)**
    • "Using causal machine learning to explore heterogeneous responses to policies"

20. **UCL Department of Statistical Sciences Seminar (January 2021)**
    • "Who benefits from health insurance? Uncovering heterogeneous policy impacts using causal machine learning"
21. **Hungarian Institute for Economic Research (Hungary 2019)**
    • "An application of causal machine learning approaches to explore heterogeneous treatment effects of social health insurance"
22. **University of Lancaster Health Economics Seminar (December 2019)**
    • "Examining effect heterogeneity using Machine Learning: the example of Health Insurance in Indonesia"
23. **MRC Biostatistics Unit 17th Armitage Workshop (November 2019)**
    • "An application of causal forests to explore the heterogeneous effects of social health insurance"
24. **University of Glasgow Institute of Health and Wellbeing: "What next for natural experimental studies?" expert workshop (November 2019)**
    • "Examining effect heterogeneity using Machine Learning: the example of Health Insurance in Indonesia"
25. **University of Newcastle Health Economics Seminar (October 2019)**
    • "An application of causal machine learning approaches to explore heterogeneous treatment effects of social health insurance"
26. **Centre for Statistical Methodology Seminar, LSHTM (September 2019)**
    • "Contrasting targeted learning and causal forests for inference about conditional average treatment effects of social health insurance programmes"
27. **University of Glasgow Maurice Bloch Seminar (June 2019)**
    • "Using machine learning in policy evaluation"
28. **University of Manchester HOPE Seminar (June 2019)**
    • "Machine learning in policy evaluation: new tools for causal Inference"
29. **INAHEA (Indonesian Health Economics Association) Conference (Indonesia, 2018) – Plenary speaker**
    • "The impact of health insurance expansion on infant mortality in Indonesia"
30. **University of California Davis, Biostatistics Seminar (USA, 2016)**
    • "Estimating the comparative effectiveness of longitudinal treatment interventions: an application in pediatric intensive care"
31. **University of California San Francisco, Biostatistics Seminar (USA, 2016)**
    • "Estimating the comparative effectiveness of longitudinal treatment interventions: an application in pediatric intensive care"
32. **Royal Statistical Society International Conference (Invited Session "Data Science: Applications of Machine Learning in Biostatistics"**
    • Estimating the comparative effectiveness of longitudinal treatment regimes: an application of targeted maximum likelihood estimation in critical care"
33. **Toronto Health Economics and Technology Assessment (THETA) Collaborative Seminar (Canada, 2015)**
34. **University College London, Department of Statistical Sciences Seminar (UK, 2015)**
35. **University of Warwick Health Economics Journal Club (UK, 2014)**
36. **Centre for Health Economics Research and Evaluation Seminar, UTS (Sydney, AU, May 2014)**
37. **University of York Centre for Health Economics Seminar (UK, 2014)**
38. **University of Manchester Econometrics Group Seminar (UK, 2013)**
39. **Health Econometrics and Data Group Seminar, University of York (UK, 2013)**
40. **Causal Inference Discussion Group Meetings at Harvard School of Public Health (USA, 2011)**

SELECTED CONTRIBUTED CONFERENCE PRESENTATIONS (POSTER OR ORAL PRESENTATION, PRESENTING AUTHOR)

1. **SMDM 48th Annual Meeting** (oral presentation in contributed session)
   - "Choosing Machine Learning Methods to Maximize Treatment Classification Accuracy in RCTs"
2. **15th Annual Conference of the Society of Health Economists (ASHEcon)** (oral presentation in organized session), Minneapolis, 2026
   - "Choosing Machine Learning Methods to Maximize Treatment Classification Accuracy in RCTs"
3. **14th Health Economics Methodology Meeting**, Vancouver 2026 (discussion paper)
   - "Choosing Machine Learning Methods to Maximize Treatment Classification Accuracy in RCTs"
4. **13th Annual Conference of the Society of Health Economists (ASHEcon)** (oral presentation in organized session), San Diego, 2024
   - "Policy learning with rare outcomes"
5. **SMDM 46th Annual Meeting** (panelist in Workshop "Decision-analytic and machine learning tools in "healthcare decision making: A debate on their acceptability in clinical practice, treatment guidelines, and health technology assessment"), Boston 2024
   - 'Informing treatment stratification using machine learning"
6. **ISPOR Europe** (**Discussant**, alongside Florent Guelfucci (discussion leader) and Imke Mayer), Barcelona, Spain (2024)
   - "Use of Machine Learning Techniques to Create External Control Arms: Guidance for Submissions to Health Technologic Assessment"
7. **ISPOR Europe** (Panelist in Workshop), Copenhagen, Denmark (2023)
   - "The role of machine learning in HTA: the academic's perspective"
8. **European Causal Inference Meeting (EuroCIM),** oral presentation, 2023
   - "Policy learning with rare outcomes"
9. **American Causal Inference Conference** (Berkeley, USA), (poster) (2022)
10. **Hungarian Economics Association conference** (oral presentation)(2021)
    - "An application of causal forests to explore the heterogeneous effects of social health insurance"
11. **International Health Economics Association (iHEA) Congress,** Basel, 2019
    - "Value for money assessment in impact evaluations"
12. **Annual Conference of the Society of Health Economists** (ASHEcon), Philadelphia (2016)
    - "Generalised synthetic controls: a critical examination of a method new to the health context" (oral presentation)
    - Discussion of the paper "Do financial penalties reduce readmission rates? A continuous treatment effect evaluation"
13. **UK-Causal Inference Meeting, London, 2016 – oral presentation**
    - "Estimating the comparative effectiveness of longitudinal treatment regimes: an illustration of targeted maximum likelihood estimation in critical care
14. **International Health Economics Association (iHEA), 2015, Milan**
    - "Health econometric evaluation of the effects of a continuous treatment: a machine learning approach"
15. **Atlantic Causal Inference Conference**, Philadelphia, 2015 – poster
16. **Atlantic Causal Inference Conference,** 2014 - poster
17. **European Health Econometrics Workshop, Munich, Germany**, 2014
    - "Health econometric evaluation of the effects of a continuous treatment: a machine learning approach"

18. **Annual Conference of the Society of Health Economists** (ASHEcon), Los Angeles (oral presentation), 2014
    - "Synthetic control method with multiple treated units: application to the Advancing Quality Program"
19. **Joint Statistical Meetings (JSM),** oral presentation**,** Montreal, 2013
20. **Society for Medical Decision Making Conference (oral presentation),** Chicago, 2012
    - "Comparison of double-robust methods for cost-effectiveness analysis"
21. **ISPOR Europe,** oral presentation in a Workshop, Madrid, 2011

SERVICE – UNIVERSITY OF WASHINGTON

- CHOICE MS Program Work Group (2026-)
- School of Pharmacy Workload task force member (2025-)
- CHOICE Curriculum Committee (2025-)
- UW CHOICE HEOR Abbvie Fellowship co-director (2024-2025)
- CHOICE PhD Student Admission Committee (2025-)
- UW ISPOR Student Chapter Faculty Advisor (2024-)
- CHOICE Data Science Journal Club faculty advisor (2024-)
- CHOICE PhD Student Coffee Chat Coordinator (2024-)

SERVICE – UNIVERSITY OF YORK

- CHE Fellowship Committee Member (2017-2022) and Chair (2022-2023)
- University of York Computing Committee Member (2017-2023)
- Line manager and academic supervisor of post-doctoral researchers
- Hiring committee member for post-doctoral research fellows

GRANT FUNDING PANEL MEMBERSHIP AND REVIEWING

- Grant funding panel membership:
    - UK Centre for Excellence and Development Impact and Learning (CEDIL) (2019);
    - UK MRC Artificial Intelligence for Better Biomedical and Health Research (2021-2024)
    - UKRI Better Methods Better Research Panel (2023-2024)
- Reviewed grant proposals for: Medical Research Council, NIHR HTA Programme, Hungarian National Research, Development and Innovation Office, College of Experts for DHSC/UKRI COVID-19 Rapid Response Rolling call, Health Foundation, Swiss National Science Foundation, University of Washington Royalty Research Fund, NASA Human Research Program, Israeli Science Foundation
- Invited reviewer for the Pedro Pita Barros Award for the Portuguese Association of Health Economics (APES)
- Reviewed grant reports for: Patient-Centered Outcomes Research Institute (PCORI)
- Abstract reviewer for conferences: Academy Health 2026, SMDM 2026, IHEA 2020, EuroCIM (2018-2025)

REFEREEING FOR PEER REVIEWED JOURNALS: *Medical Decision Making, PharmacoEconomics, Value in Health (*Selected to be Outstanding Reviewer, 2026), *Applied Health Economics and Health Policy, American Journal of Epidemiology, Epidemiology, Health Economics, Journal of Health Economics, Empirical*

Economics, Statistics in Medicine, Statistical Science, Social Science and Medicine, BMJ Open, Sociological Methods and Research, Health Policy and Planning, JRSS-A, JRSS-B, PNAS, Rapid Reviews: COVID-19, Nature Human Behavior, Econometrics Journal, Biometric Journal, Journal of Econometrics, BMC Medical Research Methodology, Data and Policy, Review of Economic Studies

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LEADING ROLE IN ORGANIZATION OF WORKSHOPS, CONFERENCES

- Scientific Committee for Society of Causal Inference Conference (2025 Detroit Conference)
- Educational Committee for Society of Causal Inference Conference (2025 -)
- Symposium on Machine Learning for Causal Inference in the Health and Social Sciences (London, UK, 2023) – main organiser
- Health Economics Study Group Meeting (York, UK) 2018 – member organising committee
- Steering group member of European Causal Inference Conference (2018 – 2024)

CHAIRING CONFERENCE SESSIONS

- Chair and discussant in the session "New frontiers in causal inference: clinical, equity and policy evaluations" at Academy Health AMR 2026, Seattle

OTHER SERVICE

- Member of The Kennedy Forum (TKF) Economics Committee [TKF is a U.S. non-profit organization focused on system-level change in mental health and substance use care]

PHD STUDENTS: MAIN ADVISOR [CHAIR] AND CO-SUPERVISOR [COMMITTEE MEMBER] ROLES

2024-2026	Ben Nwogu: "Evaluating Newer Therapies in the Treatment of Metastatic Prostate Cancer" (thesis committee member)
2024-	Emily Callen: "Evaluating the Effectiveness of State HIV Pre-exposure Prophylaxis Drug Assistance Programs" (thesis committee member)
2021-	Nadine Kraamwinkel-Jha (University of Gothenburg): "Overcoming the plateau in public health and development in low and middle-income countries: analyzing heterogeneity and effects of policies and interventions" (50% supervisor)
2019-2023	Vishalie Shah ("Machine learning methods in health policy evaluation", 50% supervisor) – thesis awarded
2019-2023	Daniela Rodrigues (Imperial College): "Evaluation of digital-first models of care using routinely collected data" (4th supervisor) – thesis awarded
2018-2022	Finn McGuire ("Methods to evaluate health policies in low-and middle income countries", 50% supervisor with Peter Smith) – thesis awarded
2017- 2021	Leonardo Koeser (King's College London): "Methods to evaluate the causal effects psychiatric interventions using observational data" (3rd Supervisor) – thesis awarded

MASTERS STUDENT: DISSERTATION ADVISOR

2026	Griffen Allen, MSc in Health Services Research (Committee co-chair)
2025-2026	Janice Jang, MSc in HEOR (Committee member, "Healthcare Utilization and Cost Burden with Switching or Cycling After Starting

	Anti-Tumor Necrosis Factors in Inflammatory Bowel Disease: A Retrospective Cohort Analysis")
2025-2026	Andrew Cotterill, MSc in HEOR (Committee chair, "Healthcare Resource Utilization and Costs Among Patients with Essential Tremor")
2024-2025	Rachel Kneitel, MSc in HEOR UW (Committee chair, "Healthcare Resource Utilization and Costs in Patients with AML Treated with Post-Transplant Maintenance Therapy")
2021	Julio Mieses Ramirez, MSc in Health Economics, U. of York
2019	Aliah Sagar and Gabriel Torres, MSc in Health Economics, U. of York
2018	Silvia Moler Zapata, MSc in Health Economics, U. of York ("Estimating the health effects of expansions in health expenditure in Indonesia: a dynamic panel data approach." Dissertation published as: Moler-Zapata et al. 2022),
2017	Sangjun Kim, MSc in Health Economics, U. of York ("From impact evaluation to decision-analysis: assessing the extent and quality of evidence on 'value for money' in health impact evaluations in low-and middle-income countries", Published as: Kreif et al. 2021)

TEACHING – MASTER AND PHD LEVEL

2026-	**CHOICE Graduate seminar** leader
2025-	Guest Lecturer (2024), Co-course-master (2025) then course master (2026-) in **Application of Machine Learning in Health Economics and Outcomes**
2011 - 2016	Graduate teaching assistant in **Introduction to Health Economics (MSc)** (at the MSc Course in Public Health, LSHTM)
2011 - 2016	Graduate teaching assistant in **Economic Evaluation (MSc)** (at LSHTM)

TEACHING – CERTIFICATE AND SHORT COURSE FACULTY

2026	**SMDM** Virtual Short Course: **Machine Learning for Causal Inference in HEOR** (November 2024)
2026	ISPOR Virtual Short Course: **Machine Learning for Causal Inference in HEOR** (June 2024)
2025	Guest lecturer (2025) then key faculty (2026-) **in the Data science in HEOR certification course,** at the University of Washington, School of Pharmacy, CHOICE Institute.
2025	Invited 2.5 hours **Workshop on Machine Learning for Health Policy Evaluation** (European Health Economics Association Early Career Researcher forum, over 100 participants)
2024	ISPOR Virtual Short Course: **Machine Learning for Causal Inference in HEOR** (July 2024)
2022	Designed and lead instructor of 4 hours short course: **Machine Learning for Causal Inference** (Health Economics Study Group Meeting, June 2022; Centre for Health Economics, July 2022, The Norwegian Research Centre (NORCE), October 2022)
2019	Short course: **Introduction to Econometric Methods for Policy and Program Evaluation** (2 lectures, Wits School of Public Health, SA)

2019	Short course: **Advanced Statistical Modelling for Economic Evaluation** (at University of York, 2 lectures and 1 computer exercises)
2018	Short course: **Performance assessment and multilevel modelling** (part the Marie Sklodowska-Curie Actions Innovative Training Networks, Hamburg, 2 lectures and 2 computer exercises)
2011 - 2016	Short course: **Methods for addressing selection bias in health economic evaluation** (run at LSHTM (2 days course) and as pre-conference 1-day short courses at ISPOR Europe)

TEACHING- UNDERGRADUATE

2007 - 2008	Instructor of **Econometrics** (undergraduate), Rajk Laszlo College for Advanced Studies, Hungary
2006	**Seminar leader of Introductory Microeconomics** (undergraduate), International Business School, Budapest